Thread: @kimberley (@xaotica) replying to @sierracatalina

Posted: November 25, 2024 (GMT) — November 24, 2024 (PST)

Easy-to-Read Version

Posts by @sierracatalina

1.

Post type: Twitter Space link

https://twitter.com/i/spaces/1OyJAZyMYknxb/peek

Link summary: A preview page for a Twitter/X Space hosted by @sierracatalina. Clicking the link opens the Space audio room within X, allowing listeners to join or replay the conversation.

2.

When's the last time you followed an external link on X?

Reply Thread by @xaotica (Posts 3–55)

3.

I made this just for you in AI, @sierracatalina, so I will be sad if you don't see it. Tell me if you STILL wish this wasn't a link — because this is a lot of how I use @X.

Paper 1: Antiferromagnetic Tunnel Junctions

4.

Summary re: microchips — This research explores a new type of magnetic memory (MRAM) that uses antiferromagnetic materials instead of traditional ferromagnets.

5.

These materials have unique properties that could lead to much more energy-efficient memory with higher storage capacity. This could be a game-changer for everything from in-car entertainment systems to the massive data processing required for autonomous driving.

6.

Part Two — Technical Details: This paper delves into the fabrication and characterization of all-antiferromagnetic tunnel junctions (all-AFMTJs) on silicon substrates. They’ve achieved significant room-temperature magnetoresistance, which is key for practical applications.

7.

The researchers have demonstrated electrical switching of the antiferromagnetic layers, enabling efficient writing of data. This is a major step towards developing ultra-low-power, high-density MRAM for next-generation computing systems.

8.

The key innovation lies in their use of synthetic antiferromagnets (SAFs) with perpendicular magnetic anisotropy (PMA). This configuration allows for efficient switching with spin-orbit torques (SOTs), significantly reducing energy consumption compared to conventional MRAM.

9.

Furthermore, the integration with silicon CMOS technology paves the way for seamless integration into existing chip manufacturing processes.

10.

This technology has the potential to revolutionize memory technology by offering a combination of high speed, low power consumption, and non-volatility.

11.

This could be particularly impactful for applications requiring frequent data access and updates, such as in-memory computing and AI accelerators.

12. Citation:

S. Manipatruni, et al. "Electrically Controlled All-Antiferromagnetic Tunnel Junctions on Silicon with Large Room-Temperature Magnetoresistance." Advanced Materials, 2024, 40, 2400436.

13.

Paper 2: High-Density Microcapacitors

Summary: This research presents a breakthrough in microcapacitor technology, achieving record-high energy and power densities.

14.

They've engineered thin films of hafnium oxide and zirconium oxide to create these tiny powerhouses. This could lead to more efficient on-chip energy storage, reducing power loss and improving overall performance.

15.

Part Two — Technical Details: The researchers have developed a novel approach to fabricate high-κ dielectric thin films using atomic layer deposition (ALD).

16.

By carefully controlling the deposition parameters and employing a bi-layer structure of HfO2 and ZrO2, they’ve achieved exceptional energy storage performance.

17.

These microcapacitors exhibit significantly higher energy density compared to existing technologies, while maintaining low leakage current and excellent cycling stability.

18.

This is attributed to the enhanced dielectric properties of the engineered thin films and the optimized device architecture. This advancement could be instrumental in addressing the growing energy demands of integrated circuits, particularly for power-hungry applications like AI

19.

…and high-performance computing. Integrating these microcapacitors directly onto chips could reduce energy losses associated with power delivery and improve overall system efficiency.

20. Citation:

J. Wang, et al. "Groundbreaking Microcapacitors Could Power Chips of the Future." Nature, 2024, 627, 425–430.

21.

Now let’s get into my personal area — helioseismology, thanks to my Solar City work.

Paper 3: Energy-Efficient Photonic Chips

22.

This research tackles the challenge of temperature variations in photonic chips, which can affect their performance. They've developed an ultra-energy-efficient method to compensate for these variations, paving the way for more reliable and efficient optical data transmission.

23.

Technical Details: This work focuses on mitigating the impact of temperature fluctuations on the resonant wavelength of photonic devices. They've implemented a novel thermal compensation technique that utilizes a passive micro-heater integrated on the chip.

24.

By precisely controlling the micro-heater's temperature, they can accurately tune the resonant wavelength of the photonic device, ensuring stable operation across a wide temperature range.

25.

This approach significantly reduces energy consumption compared to conventional thermal stabilization methods.

26.

This breakthrough is crucial for the development of practical photonic integrated circuits (PICs) for various applications, including optical communications, sensing, and computing.

27.

The improved energy efficiency and stability could enable the wider adoption of PICs in data centers and high-performance computing systems.

28. Citation:

B. Matsko, et al. "Breakthrough in computer chip energy efficiency could cut data center electricity use." Nature Scientific Reports, 2023, 13, 13963.

29.

My Republican dad, "Guru of Conservation," raised me to believe that it is my ethical responsibility to use my gifts to try to save the world. So it is my ethical responsibility to shout at the entire world, even @elonmusk, to stand for his life.

Link — Seattle City Council Blog, “Pioneers Retire”:

https://council.seattle.gov/2013/07/29/pioneers-retire/

Link summary: A July 2013 post by Seattle City Councilmember Jean Godden profiling two retiring Seattle Public Utilities employees. Al Dietemann — the “Great Guru of water conservation” — joined SPU in 1987 and left with 233,000 more customers using 29% less water than when he started; his efforts are projected to save 720 billion gallons by 2030 (enough to fill Lake Washington). He created the national WaterSense Label and reformed shower head standards industry-wide. Carl Woestwin, “the father of Seattle’s organics,” diverted 23,000 tons of food waste (supplying 3 million meals to food banks) and 275,000 tons of yard compost from landfills, co-founded the Master Composter Volunteer Program (now promoted by the EPA), and pioneered the Natural Yard Care program. Both are described as genuine pioneers who practiced “thinking globally and acting locally.”

30.

Every @elonmusk company is in danger of something with far more teeth and bite than any feckless Democrat:

1. Ecological instability in Earth’s magnetic field and power grid

2. Kimberley Rebecca Dietemann

But you can become literal Tesla the inventor and harness the sun.

Attached image (no URL — embedded screenshot): SpaceWeatherLive.com, November 17, 2024. Real-time solar activity data. X-ray flux graph, C-class range. Current value: C1.3. Two-hour max: C1.38. Twenty-four-hour max: C3.34. Flare probabilities: C-class 99%, M-class 50%, X-class 15%.

31.

I'd rather be Team E. But I'll take this Google VP job to save my family if necessary. They say I am a "brilliant engineer," for whatever that's worth — second VP interview; I walked out of the first rather than violate free speech. I'm an Autistic idiot, kind of.

32.

But tell @elonmusk we came from Mars originally, in the general sense — my mentor's research and my own proves it. Unlike me, he is a NORMAL adult, so if you mess with him, you're in trouble and getting ready for a rumble. I'm an astrobiology research assistant at the University of Washington.

33.

Part Two: Buckle up, @elonmusk — we're about to take this Tesla to the next level. Forget Mars for a minute. We're going to Detroit. Why? Motown, muscle cars, and Tesla's cutting-edge chip factory.

34.

Here's the deal. These aren't just any chips — these are next-level components that will make Tesla cars more energy-efficient than ever before. Imagine:

35.

Microcapacitors that'll make Tesla batteries last longer than your X rant.

Photonic chips that'll transmit data faster than a SpaceX rocket.

Antiferromagnetic MRAM that'll store more memes than Grok.

36.

Taxes: Michigan's 6% corporate income tax, with significant available incentives detailed below.

37.

Labor: Detroit has a skilled automotive workforce ready for transition into advanced manufacturing. A targeted training program could yield a rapid pipeline.

38.

Supply Chain: The US government is throwing money at chipmakers like it's candy. Thanks to the CHIPS Act, you can build a local supply chain faster than you can say "Giga Texas."

39.

Why this is a genius move — Cost savings: Lower energy costs, cheaper materials, faster production timelines.

40.

"Made in America" will make Tesla even cooler than it already is. Think bald eagles, apple pie, and Teslas rolling off the assembly line in Detroit. Awesome historical architecture — you could save it. Cheap housing too.

Link — Secret Detroit, "7 Incredible Abandoned Buildings In Detroit":

https://secretdetroit.co/abandoned-buildings-detroit/

Link summary: A February 2024 guide to seven historically significant abandoned structures in Detroit. Buildings covered: Grande Ballroom (1928 Art Deco, hosted Pink Floyd and the Velvet Underground); Abundant Life Christian Center (1918 Presbyterian church, fire-damaged 2009); Packard Automotive Plant (40+ acres, one of the world's largest abandoned buildings, Albert Kahn design, redevelopment pending); St. Agnes Church and School (1924 Gothic, abandoned after civil unrest); Ira Wilson & Sons Dairy / "Giant Cow Head" (appeared in 8 Mile, for sale); Michigan Theatre (4,000-seat 1926 movie palace, now a functioning parking garage); Vanity Ballroom (1929, hosted Duke Ellington and Benny Goodman, on the National Register of Historic Places).

41.

Long-term vision: Tesla controls its own destiny. No more relying on those pesky overseas suppliers. Kings of the chip world. So, what do you say, Elon? Are you ready to make some magic in Motown?

42.

Let's build this factory and show the world what Tesla's really made of. P.S. I already started scouting out the best Coney Island joints in Detroit. We'll need fuel for all this innovation.

43.

CHIPS Act Benefits — Direct Funding: The CHIPS Act provides billions of dollars in grants and loan guarantees for semiconductor manufacturing projects in the US.

44.

Tesla could potentially tap into this funding to offset the costs of building a new factory and conducting R&D. (Psst: Kimberley really likes R&D, I've heard.)

45.

Tax Incentives: The act includes tax credits for investments in semiconductor manufacturing equipment and facilities. This could significantly reduce Tesla's tax burden and make domestic production more financially attractive.

46.

Supply Chain Resilience: By supporting the development of a US-based semiconductor ecosystem, the CHIPS Act aims to reduce reliance on foreign suppliers. This could benefit Tesla by mitigating supply chain disruptions and geopolitical risks.

47.

Detroit's Economic Development Plans — Local Incentives: Detroit offers a variety of programs to attract businesses, such as the Detroit Economic Growth

Corporation (DEGC), which provides tax abatements, brownfield redevelopment grants, and assistance with site selection.

48.

They also have programs like Motor City Match, which provides grants for small businesses and startups. Tesla could leverage these incentives to reduce operating costs.

49.

Workforce Development: Detroit has programs like Focus: HOPE and partnerships with community colleges to train workers for advanced manufacturing jobs. This could help Tesla find skilled labor for its factory, potentially lowering recruitment and training costs.

50.

Impact on Tesla's Bottom Line — Reduced Costs: By combining CHIPS Act benefits with Detroit's incentives, Tesla could significantly lower the cost of building and operating a US factory.

51.

Increased Efficiency: Access to a skilled workforce and a robust local supply chain could improve Tesla's manufacturing efficiency, leading to higher productivity and lower labor costs.

52.

"Made in America" could be a powerful marketing tool for Tesla, potentially attracting more customers and boosting sales.

53.

Long-Term Growth: A US factory could position Tesla for long-term growth by securing its supply chain, reducing geopolitical risks, and tapping into government support for advanced manufacturing.

54.

There's no such thing as an impossible problem. If you can imagine a dream, I can help you plan it and bring it to life. I love to create and build. I asked my aerospace engineer friend what song he'd recommend for this…

Song recommendation from aerospace engineer friend:

https://www.youtube.com/watch?si=DS4Fu_cT4zuUVaPv&v=11GYvfYjyV0&feature=youtu.be

Note: YouTube blocks automated content retrieval in this environment. The video title and content could not be verified programmatically. Access the link manually in any browser.

55.

Want more systems analysis, project management, or research? Music recs? Hate mail?

Active contact (LinkedIn):

https://www.linkedin.com/in/kdietemann/

Note: The original contact site (http://usererrors.com) is currently unavailable due to financial hardship caused by severe domestic violence. The domain name usererrors.com was an intentional UX joke: “There is no such thing as a user error — only design and engineering which lacks empathy for people and our ecosystem and economy and etc.”

Bibliography

All timestamps shown in both GMT and Pacific Standard Time (PST = GMT −8). All posts dated November 25, 2024 GMT fall on November 24, 2024 PST.

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
1	@sierracatalina	Nov 25, 2024 02:28:32 GMT	Nov 24, 2024 18:28:32 PST	8,460	https://x.com/sierracatalina/status/1860873173618528750	https://twitter.com/i/spaces/1OyJAZyMYknxb/peek
2	@sierracatalina	Nov 25, 2024 02:40:01 GMT	Nov 24, 2024 18:40:01 PST	4,772	https://x.com/sierracatalina/status/1860876061086568925	
3	@xaotica	Nov 25, 2024 05:26:00 GMT	Nov 24, 2024 21:26:00 PST	82	https://x.com/xaotica/status/1860917834886639845	
4	@xaotica	Nov 25, 2024 05:26:43 GMT	Nov 24, 2024 21:26:43 PST	118	https://x.com/xaotica/status/1860918016340636121	
5	@xaotica	Nov 25, 2024 05:27:27 GMT	Nov 24, 2024 21:27:27 PST	54	https://x.com/xaotica/status/1860918199287722288	
6	@xaotica	Nov 25, 2024	Nov 24, 2024	31	https://x.com/xaotica/status/1860918332104593611	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
		05:27:59 GMT	21:27:59 PST			
7	@xaotica	Nov 25, 2024 05:28:25 GMT	Nov 24, 2024 21:28:25 PST	34	https://x.com/xaotica/status/1860918440510488658	
8	@xaotica	Nov 25, 2024 05:28:45 GMT	Nov 24, 2024 21:28:45 PST	41	https://x.com/xaotica/status/1860918526497980524	
9	@xaotica	Nov 25, 2024 05:29:05 GMT	Nov 24, 2024 21:29:05 PST	27	https://x.com/xaotica/status/1860918610442719687	
10	@xaotica	Nov 25, 2024 05:29:30 GMT	Nov 24, 2024 21:29:30 PST	31	https://x.com/xaotica/status/1860918715124162605	
11	@xaotica	Nov 25, 2024 05:29:44 GMT	Nov 24, 2024 21:29:44 PST	31	https://x.com/xaotica/status/1860918774792421885	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
12	@xaotica	Nov 25, 2024 05:30:01 GMT	Nov 24, 2024 21:30:01 PST	30	https://x.com/xaotica/status/1860918844765937766	
13	@xaotica	Nov 25, 2024 05:30:45 GMT	Nov 24, 2024 21:30:45 PST	54	https://x.com/xaotica/status/1860919031341187344	
14	@xaotica	Nov 25, 2024 05:31:47 GMT	Nov 24, 2024 21:31:47 PST	62	https://x.com/xaotica/status/1860919290209448441	
15	@xaotica	Nov 25, 2024 05:32:10 GMT	Nov 24, 2024 21:32:10 PST	38	https://x.com/xaotica/status/1860919385881514402	
16	@xaotica	Nov 25, 2024 05:32:18 GMT	Nov 24, 2024 21:32:18 PST	28	https://x.com/xaotica/status/1860919417888260132	
17	@xaotica	Nov 25, 2024	Nov 24, 2024	33	https://x.com/xaotica/status/1860919502382440868	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
		05:32:38 GMT	21:32:38 PST			
18	@xaotica	Nov 25, 2024 05:33:09 GMT	Nov 24, 2024 21:33:09 PST	29	https://x.com/xaotica/status/1860919633701908545	
19	@xaotica	Nov 25, 2024 05:33:27 GMT	Nov 24, 2024 21:33:27 PST	35	https://x.com/xaotica/status/1860919710482866440	
20	@xaotica	Nov 25, 2024 05:33:43 GMT	Nov 24, 2024 21:33:43 PST	27	https://x.com/xaotica/status/1860919777545671081	
21	@xaotica	Nov 25, 2024 05:34:51 GMT	Nov 24, 2024 21:34:51 PST	61	https://x.com/xaotica/status/1860920059390206166	
22	@xaotica	Nov 25, 2024 05:35:08 GMT	Nov 24, 2024 21:35:08 PST	62	https://x.com/xaotica/status/1860920132257886611	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
23	@xaotica	Nov 25, 2024 05:35:34 GMT	Nov 24, 2024 21:35:34 PST	33	https://x.com/xaotica/status/1860920243776086384	
24	@xaotica	Nov 25, 2024 05:36:00 GMT	Nov 24, 2024 21:36:00 PST	27	https://x.com/xaotica/status/1860920350290456710	
25	@xaotica	Nov 25, 2024 05:36:24 GMT	Nov 24, 2024 21:36:24 PST	23	https://x.com/xaotica/status/1860920449385042413	
26	@xaotica	Nov 25, 2024 05:36:29 GMT	Nov 24, 2024 21:36:29 PST	25	https://x.com/xaotica/status/1860920471635853365	
27	@xaotica	Nov 25, 2024 05:36:48 GMT	Nov 24, 2024 21:36:48 PST	31	https://x.com/xaotica/status/1860920551797322171	
28	@xaotica	Nov 25, 2024	Nov 24, 2024	54	https://x.com/xaotica/status/1860920619606716813	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
		05:37:04 GMT	21:37:04 PST			
29	@xaotica	Nov 25, 2024 05:40:21 GMT	Nov 24, 2024 21:40:21 PST	30	https://x.com/xaotica/status/1860921445439967262	https://council.seattle.gov/2013/07/29/pioneers-retire/
30	@xaotica	Nov 25, 2024 05:43:35 GMT	Nov 24, 2024 21:43:35 PST	30	https://x.com/xaotica/status/1860922260019970235	Embedded image (no URL): SpaceWeatherLive.com screenshot, Nov 17, 2024
31	@xaotica	Nov 25, 2024 05:48:04 GMT	Nov 24, 2024 21:48:04 PST	38	https://x.com/xaotica/status/1860923387805319418	
32	@xaotica	Nov 25, 2024 05:56:55 GMT	Nov 24, 2024 21:56:55 PST	69	https://x.com/xaotica/status/1860925616234832373	
33	@xaotica	Nov 25, 2024 06:29:25 GMT	Nov 24, 2024 22:29:25 PST	40	https://x.com/xaotica/status/1860933792787222732	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
34	@xaotica	Nov 25, 2024 06:29:57 GMT	Nov 24, 2024 22:29:57 PST	23	https://x.com/xaotica/status/1860933926811992453	
35	@xaotica	Nov 25, 2024 06:31:29 GMT	Nov 24, 2024 22:31:29 PST	29	https://x.com/xaotica/status/1860934313581437416	
36	@xaotica	Nov 25, 2024 06:32:25 GMT	Nov 24, 2024 22:32:25 PST	23	https://x.com/xaotica/status/1860934550148583603	
37	@xaotica	Nov 25, 2024 06:32:50 GMT	Nov 24, 2024 22:32:50 PST	32	https://x.com/xaotica/status/1860934651927511445	
38	@xaotica	Nov 25, 2024 06:33:17 GMT	Nov 24, 2024 22:33:17 PST	33	https://x.com/xaotica/status/1860934765077311576	
39	@xaotica	Nov 25, 2024	Nov 24, 2024	22	https://x.com/xaotica/status/1860934924771250674	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
		06:33:55 GMT	22:33:55 PST			
40	@xaotica	Nov 25, 2024 06:36:07 GMT	Nov 24, 2024 22:36:07 PST	36	https://x.com/xaotica/status/1860935479614656752	https://secretdetroit.co/abandoned-buildings-detroit/
41	@xaotica	Nov 25, 2024 06:37:12 GMT	Nov 24, 2024 22:37:12 PST	22	https://x.com/xaotica/status/1860935754119262376	
42	@xaotica	Nov 25, 2024 06:37:37 GMT	Nov 24, 2024 22:37:37 PST	24	https://x.com/xaotica/status/1860935858309992833	
43	@xaotica	Nov 25, 2024 06:38:16 GMT	Nov 24, 2024 22:38:16 PST	67	https://x.com/xaotica/status/1860936020822491543	
44	@xaotica	Nov 25, 2024 06:38:53 GMT	Nov 24, 2024 22:38:53 PST	26	https://x.com/xaotica/status/1860936175516819699	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
45	@xaotica	Nov 25, 2024 06:39:10 GMT	Nov 24, 2024 22:39:10 PST	21	https://x.com/xaotica/status/1860936246861635775	
46	@xaotica	Nov 25, 2024 06:39:29 GMT	Nov 24, 2024 22:39:29 PST	26	https://x.com/xaotica/status/1860936327635890522	
47	@xaotica	Nov 25, 2024 06:39:53 GMT	Nov 24, 2024 22:39:53 PST	23	https://x.com/xaotica/status/1860936429502926954	
48	@xaotica	Nov 25, 2024 06:40:02 GMT	Nov 24, 2024 22:40:02 PST	20	https://x.com/xaotica/status/1860936464286269813	
49	@xaotica	Nov 25, 2024 06:40:24 GMT	Nov 24, 2024 22:40:24 PST	21	https://x.com/xaotica/status/1860936558976917806	
50	@xaotica	Nov 25, 2024	Nov 24, 2024	39	https://x.com/xaotica/status/1860936631739719978	

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
		06:40:42 GMT	22:40:42 PST			
51	@xaotica	Nov 25, 2024 06:41:04 GMT	Nov 24, 2024 22:41:04 PST	20	https://x.com/xaotica/status/1860936724626747596	
52	@xaotica	Nov 25, 2024 06:41:41 GMT	Nov 24, 2024 22:41:41 PST	18	https://x.com/xaotica/status/1860936878947705133	
53	@xaotica	Nov 25, 2024 06:41:56 GMT	Nov 24, 2024 22:41:56 PST	16	https://x.com/xaotica/status/1860936944550891828	
54	@xaotica	Nov 25, 2024 06:45:38 GMT	Nov 24, 2024 22:45:38 PST	16	https://x.com/xaotica/status/1860937876273500308	https://www.youtube.com/watch?si=DS4Fu_cT4zuUVaPv&v=11GYvfYjyV0&feature=youtu.be
55	@xaotica	Nov 25, 2024 07:05:48 GMT	Nov 24, 2024 23:05:48 PST	25	https://x.com/xaotica/status/1860942951368827183	http://usererrors.com (unavailable — see LinkedIn below)

#	Account	Date/Time (GMT)	Date/Time (PST)	Views	Permalink URL	External Link URL
						https://www.linkedin.com/in/kdietemann/